

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	AB Dhanyashree	Reg. No.:	192472002
Section / Batch:	D/2024	Date:	16.7.26

Code of Conduct

I AB Dhanyashree(192472002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: AB

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Annexure A – Pattern Recognition and Text Processing

Section 1: Regular Expression Pattern Generation

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

For the given student details formulate Regex patterns for the following:

Email ID

Mobile Number

Strong Password

Date of Birth (DD/MM/YYYY)

Register Number

Section 2: Search for a String

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

Write a Python program using the re module to perform the following operations for the word "AI":

1. Find the first occurrence of the word "AI" using re.search().
2. Display the starting position of the first occurrence.
3. Display the ending position of the first occurrence.
4. Display the total number of times the word "AI" appears in the passage.
5. Print an appropriate message if the word is not found.

Section 3: Split the documentation into sentences and words

Refer the previous passage and Write a Python program using the re.split() method to perform the following operations:

- Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.
- Count and display the total number of sentences.
- Split the passage into words by considering one or more whitespace characters as delimiters.
- Count and display the total number of words.
- Display the list of extracted sentences and words.

Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

Email ID

- Must begin with one or more letters or digits.
- May contain ., %, +, or - before the @ symbol.
- Must contain a valid domain name.
- Must end with a valid domain extension such as .com, .org, .edu, or .in.

Strong Password

- Must contain at least 8 characters.
- Must contain at least one uppercase letter.
- Must contain at least one lowercase letter.
- Must contain at least one digit.
- Must contain at least one special character from @, #, \$, %, &, !, or *.
- Must not contain any whitespace characters.

Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the re.compile() method to compile the pattern once and reuse it to validate all the email IDs.

Question 1:

For the given student details formulate Regex patterns for the following:

Email ID

```
^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$
```

Mobile Number

```
^(\+91[- ])?[6-9]\d{9}$
```

Strong Password

```
^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%&*!])[A-Za-z\d@#%&*!]{8,}$
```

Date of Birth (DD/MM/YYYY)

```
^(0[1-9]|[12][0-9]|3[01])/(0[1-9]|1[0-2])/(19|20)\d{2}$
```

Register Number

```
^\d{2}[A-Z]{4}\d{4}$
```

Question 2:

Coding:

```
import re
```

```
text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences.
```

```
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.
```

```
As AI continues to evolve, professionals with AI skills are in high demand.
```

```
"""
```

```
match = re.search(r"\bAI\b", text)
```

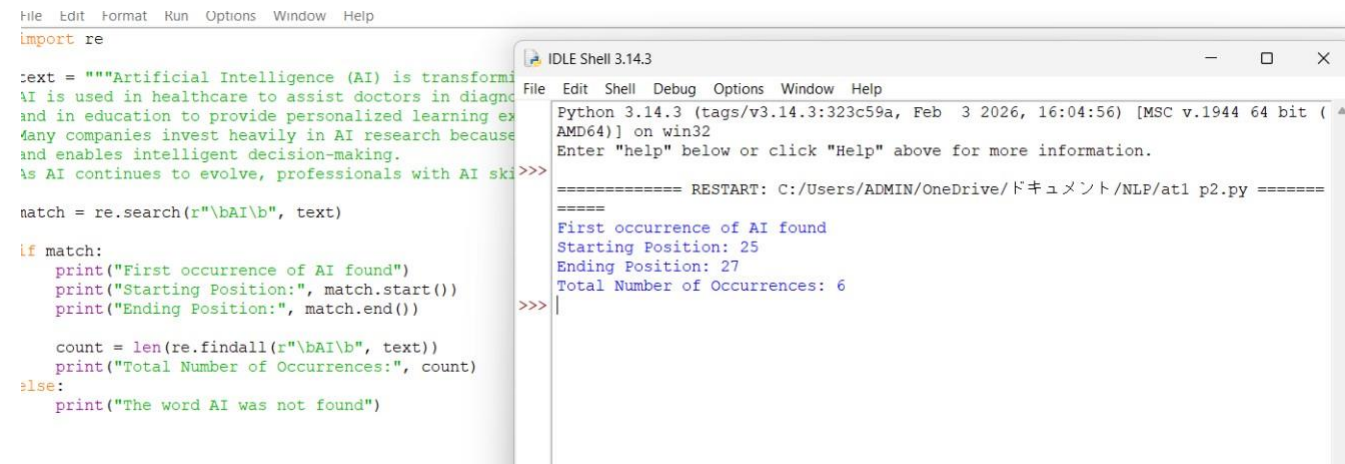
```
if match:
```

```
    print("First occurrence of 'AI' found.")
    print("Starting Position:", match.start())
```

```
print("Ending Position:", match.end())
```

```
occurrences = re.findall(r"\bAI\b", text) print("Total  
Number of Occurrences:", len(occurrences)) else:  
print("The word 'AI' was not found.")
```

OUTPUT:

A screenshot of a Python IDE window titled 'IDLE Shell 3.14.3'. The left pane shows Python code for finding the first occurrence and total count of 'AI' in a text string. The right pane shows the output of the code execution. The code defines a text string about AI, searches for the first 'AI' using re.search, and prints its starting and ending positions. It also counts all occurrences of 'AI' using re.findall and prints the total count. The output shows the first occurrence at starting position 25 and ending position 27, and a total of 6 occurrences.

```
File Edit Format Run Options Window Help
import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent
decision-making.
As AI continues to evolve, professionals with AI skills are in high demand."""

match = re.search(r"\bAI\b", text)

if match:
    print("First occurrence of AI found")
    print("Starting Position:", match.start())
    print("Ending Position:", match.end())

    count = len(re.findall(r"\bAI\b", text))
    print("Total Number of Occurrences:", count)
else:
    print("The word AI was not found")

IDLE Shell 3.14.3
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ADMIN/OneDrive/ドキュメント/NLP/at1 p2.py =====
First occurrence of AI found
Starting Position: 25
Ending Position: 27
Total Number of Occurrences: 6
>>>
```

Question 3:

Coding:

```
import re
```

```
text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent
decision-making.
As AI continues to evolve, professionals with AI skills are in high demand."""
```

```
sentences = re.split(r'[.!?]+', text)
```

```
sentences = [s.strip() for s in sentences if s.strip()]
```

```
print("Sentences:") for
s in sentences:
    print(s)
```

```
print("\nTotal Number of Sentences:", len(sentences))
```

```
words = re.split(r'\s+', text.strip())
```

```
print("\nWords:")
```

```
print(words)
```

```
print("\nTotal Number of Words:", len(words))
```

Output:

```
>>>
===== RESTART: C:/Users/ADMIN/OneDrive/ドキュメント/NLP/at1 p3.py =====
Sentences:
Artificial Intelligence (AI) is transforming industries across the world
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making
As AI continues to evolve, professionals with AI skills are in high demand

Total Number of Sentences: 4

Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis', 'in', 'banking', 'to', 'detect', 'fraud,', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand.']

Total Number of Words: 60
>>>
```

Question 4:

Coding:

```
import re
```

```
email = input("Enter Email ID: ")
```

```
password = input("Enter Password: ")
```

```
email_pattern = r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
```

```
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$$%&!*])[A-Za-z\d@#$$%&!*]{8,}$'
```

```
if re.fullmatch(email_pattern, email):
```

```
    print("Valid Email ID") else:
```

```
    print("Invalid Email ID")
```

```
if re.fullmatch(password_pattern, password):
```

```
    print("Strong Password") else:
```

```
    print("Invalid Password")
```

Output:



The image shows a Python script in an IDE window titled 'at1 p4.py' and its execution output in an 'IDLE Shell 3.14.3' window. The script defines two regular expressions for email and password validation. The email pattern is `r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'` and the password pattern is `r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[!@#$%^&*~`|;:~\'])\S+$'`. The script prompts the user to enter an email ID and a password. The shell window shows the output of the script, including the restart message, the email input 'Vimala.kvv123@gmail.com', the password input '12vimala@34', and the validation results: 'Valid Email ID' and 'Invalid Password'.

```
at1 p4.py - C:/Users/ADMIN/OneDrive/ドキュメント/NLP/at1 p4.py (3.14.3)
File Edit Format Run Options Window Help

import re

# Get user input
email = input("Enter Email ID: ")
password = input("Enter Password: ")

# Regex for Email ID
email_pattern = r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'

# Regex for Strong Password
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[!@#$%^&*~`|;:~\'])\S+$'

# Validate Email
if re.fullmatch(email_pattern, email):
    print("Valid Email ID")
else:
    print("Invalid Email ID")

# Validate Password
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Invalid Password")

IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ADMIN/OneDrive/ドキュメント/NLP/at1 p4.py =====
>>>
Enter Email ID: Vimala.kvv123@gmail.com
Enter Password: 12vimala@34
Valid Email ID
Invalid Password
>>>
```

Question 5

Coding:

```
import re
email_pattern = re.compile(r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$')

emails = [
    "dhanya.kvv@gmail.com",
    "student1624@college.edu",
    "dhanya@yahoo.in",
    "dhanu@company.org",
    "invalid@email",
    "@gmail.com",
    "test123@gmail.com"
]

for email in emails:
    if email_pattern.fullmatch(email):
        print(email, "-> Valid Email")
    else:
        print(email, "-> Invalid Email")
```

Output:

```
dd - C:/Users/Dhanyashree/dd (3.10.11)
File Edit Format Run Options Window Help

import re
email_pattern = re.compile(
    r'^[A-Za-z0-9]+[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
)

emails = [
    "dhanya.kvv@gmail.com",
    "student1624@college.edu",
    "dhanya@yahoo.in",
    "dhanu@company.org",
    "invalid@email",
    "@gmail.com",
    "test123@gmail.com"
]

for email in emails:
    if email_pattern.fullmatch(email):
        print(email, "-> Valid Email")
    else:
        print(email, "-> Invalid Email")
```

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help

Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
===== RESTART: C:/Users/Dhanyashree/dd =====
dhanya.kvv@gmail.com -> Valid Email
student1624@college.edu -> Valid Email
dhanya@yahoo.in -> Valid Email
dhanu@company.org -> Valid Email
invalid@email -> Invalid Email
@gmail.com -> Invalid Email
test123@gmail.com -> Valid Email

>>>
```

Ln: 12 Col: 0

Ln: 15 Col: 0

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1				4		20	97
2				4		20	
3				4		19	
4				3		19	
5						19	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____